

Objective:

Analyze customer and sales data to identify purchasing trends, improve data quality, and create interactive dashboards that support business decision making.

Data Challenges

- Cleaned and standardized inconsistent retail data using SQL by handling duplicate records, missing values, formatting inconsistencies, invalid entries, and inconsistent category labels to ensure accurate analysis and dashboard reporting.

Key Insights

- Customers aged 26-35 and 36-50 generate 31.14% and 31.49% of the total revenue and has highest order volume, indicating strong and consistent customer activity, while customers aged 18-25 records the highest average order value (26.26K), suggesting higher value purchases despite lower transaction frequency.
- Male customers form the majority revenue of 57.2% and place slightly more orders (51.3%), while female customers remain nearly as engaged as male.
- Female customers have larger wishlist and show a higher return rate.
- The primary customer acquisition channels are Organic Search, Google Ads, and Facebook Ads, with Organic and Google Ads generating the highest revenue.
- Electronics and Fashion are the most preferred product categories across both genders.
- Customers from Tier 1 cities generate the highest revenue, while Tier 3 cities experience higher return rates.
- The top selling product categories are Electronics, Fashion, Clothing, Grocery, and Home products, while Gems, Personal Care, Luxury items, Furniture, and Premium products show lower sales volume.
- Grocery, Apparel, Food, Clothing, and Beauty products receive the highest discount usage.
- Long term customer loyalty strongly impacts revenue growth. Customers with older accounts spend more, return fewer items, and demonstrate more stable purchasing behavior, while newer users interact more frequently with the platform but spend less overall.
- Home Decor, Jewelry, Luxury, and Gems generate the highest average spending per purchase, while Electronics, Fashion, and Clothing drive the highest total revenue due to larger purchase volumes.

- Customer segmentation analysis shows 2.15% are likely to churn.

Recommendations

- Focus marketing and loyalty campaigns on customers aged 26–50.
- Promote premium and luxury products to customers aged 18–25.
- Increase investment in Organic Search and Google Ads, as they are the strongest revenue generating customer acquisition channels.
- Improve retention and reduce return rates by enhancing customer experience, especially for female customers and customers from Tier 3 cities.